



Grade 7 FAST Mathematics

2026 Released Test Items

1. An equation of a proportional relationship is shown.

$$y = 2x$$

Which table represents the relationship?

Ⓐ

x	0	2	4
y	0	1	2

Ⓒ

x	0	1	2
y	2	4	6

Ⓑ

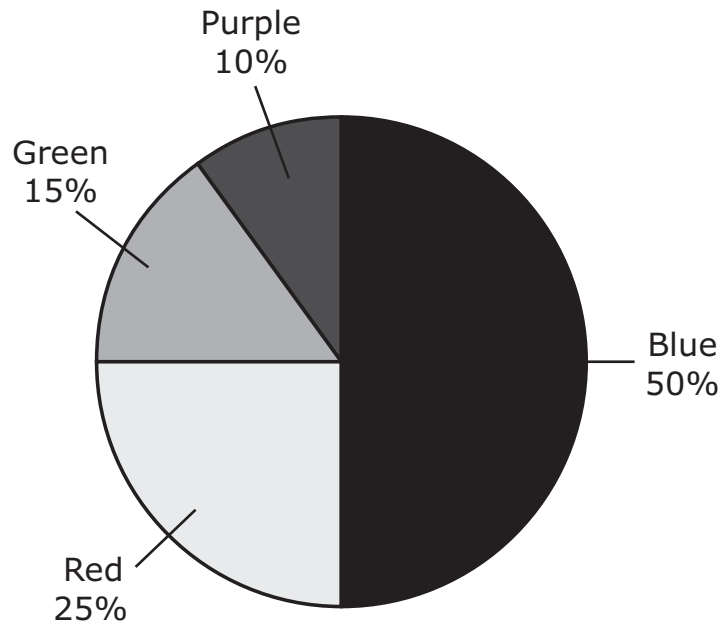
x	0	1	2
y	0	2	4

Ⓓ

x	0	1	2
y	2	4	8

2. In Xander's school club, students vote for their favorite colors. The students' data are shown in the circle graph.

Students' Favorite Colors

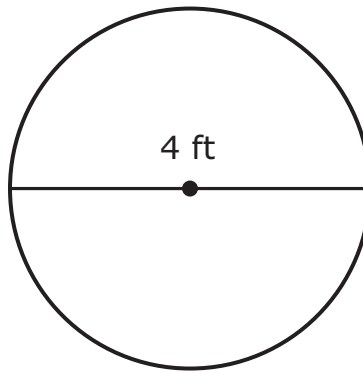


Xander knows that 6 students voted for green.

What is the total number of students who voted?

- Ⓐ 15
- Ⓑ 20
- Ⓒ 34
- Ⓓ 40

3. A circle has a diameter of 4 feet (ft), as shown.



What is the circumference, in feet, of the circle? Round your answer to the nearest tenth.

←	→	↶	↷	✖	✓
1	2	3			
4	5	6			
7	8	9			
	0				
.	-	$\frac{\square}{\square}$			

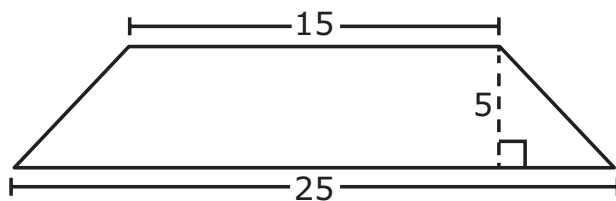
4. An expression is shown.

$$\frac{(7^3)^2 \cdot (3^2)^3}{(3 \cdot 7)^4 \cdot 7^2}$$

What is the value of the expression?

- Ⓐ $\frac{3}{7}$
- Ⓑ 9
- Ⓒ $\frac{81}{7}$
- Ⓓ 63

5. A trapezoid, with dimensions in units, is shown.



What is the area, in square units, of the trapezoid?

- Ⓐ 50
- Ⓑ 100
- Ⓒ 125
- Ⓓ 200

Grade 7 FAST Mathematics

6. Liam rolls a fair 10-sided die. The faces are numbered 1 through 10.

What is the theoretical probability that Liam rolls an 8 or higher? Express your answer as a fraction or a decimal.

←

→

↶

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✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

7. An inequality is shown.

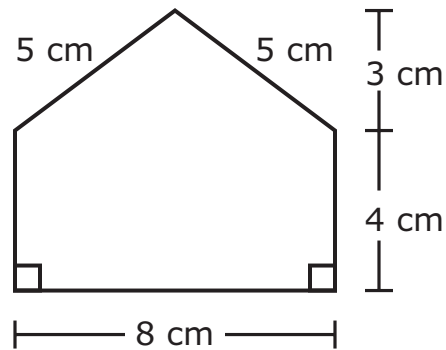
$$6 < x + 4$$

What is the solution to the inequality?

- Ⓐ $x > 2$
- Ⓑ $x < 2$
- Ⓒ $x > 10$
- Ⓓ $x < 10$

Grade 7 FAST Mathematics

8. Jada makes gingerbread cookies in the shape of a house. The dimensions, in centimeters (cm), of one cookie are shown.



What is the area, in square centimeters, of one cookie?

- Ⓐ 39.5
- Ⓑ 44
- Ⓒ 52
- Ⓓ 56

9. A spinner has five sections that are each a different color. The probability of the spinner arrow stopping on each of the colors is shown.

$$\begin{aligned}P(\text{blue}) &= 0.1 \\P(\text{green}) &= 0.19 \\P(\text{purple}) &= 0.03 \\P(\text{red}) &= 0.18 \\P(\text{yellow}) &= 0.5\end{aligned}$$

Which statement is true?

- Ⓐ It is least likely for the arrow to land on blue.
- Ⓑ It is less likely for the arrow to land on red than on blue.
- Ⓒ It is more likely for the arrow to land on blue than on green.
- Ⓓ It is equally likely and unlikely for the arrow to land on yellow.

- 10.** A hotel owner increases the price of a room from \$139 per night to \$174 per night.

By about what percentage did the hotel owner increase the price of the room?

- Ⓐ 20%
- Ⓑ 25%
- Ⓒ 35%
- Ⓓ 50%

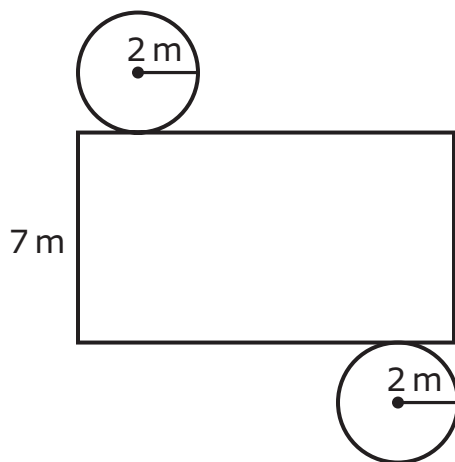
- 11.** An architect creates a drawing in which a rectangular deck is 11 inches long and 6 inches wide. The drawing is at a scale of 1 inch = 2 feet.

What is the actual area, in square feet, of the deck?

←
→
↶
↷
✖
✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

12. The net of a right circular cylinder is shown, with a radius of 2 meters (m) and a height of 7 meters.



Which expression can be used to find the surface area, in square meters, of the cylinder?

- Ⓐ $\pi(2)^2 + 2\pi(2)(7)$
- Ⓑ $\pi(4)^2 + 2\pi(4)(7)$
- Ⓒ $2\pi(2)^2 + 2\pi(2)(7)$
- Ⓓ $2\pi(4)^2 + 2\pi(4)(7)$

13. Which expression is equivalent to $2.4(7.5 + 10p) - 15p$?

- Ⓐ $27p$
- Ⓑ $18 - 5p$
- Ⓒ $18 + 9p$
- Ⓓ $18 - 12p$

Grade 7 FAST Mathematics

14. Jon's music player picks 100 songs at random. He records the genres of the songs in the table shown.

Genre	Number of Songs
Country	23
Hip-Hop	13
Pop	27
Rock	37

Based on the table, what is the experimental probability that the music player will pick a rock song? Express your answer as a fraction or a decimal.

←

→

↶

↷

✖

✓

1

2

3

4

5

6

7

8

9

0

.

-

$\frac{\square}{\square}$

15. Six more than one-half of x is equal to 16.

Which equation models this description?

Ⓐ $\frac{1}{2}x + 6 = 16$

Ⓑ $6x + \frac{1}{2} = 16$

Ⓒ $\left(\frac{1}{2} + 6\right)x = 16$

Ⓓ $6\left(x + \frac{1}{2}\right) = 16$

16. Marcus and Shelly both rewrite $1.\bar{5}$ as a fraction.

- Marcus claims $1.\bar{5} = \frac{15}{10}$ because $1.\bar{5} = 1 + \frac{5}{10} = \frac{15}{10}$.
- Shelly claims $1.\bar{5} = \frac{14}{9}$ because $1.\bar{5} = 1 + \frac{5}{9} = \frac{14}{9}$.

Which statement describes who is correct?

- Ⓐ Shelly is correct because $0.\bar{5}$ is equivalent to $\frac{5}{9}$.
- Ⓑ Marcus is correct because $0.\bar{5}$ is equivalent to $\frac{5}{10}$.
- Ⓒ Neither Marcus nor Shelly is correct because $0.\bar{5}$ is not equivalent to $\frac{5}{10}$ or $\frac{5}{9}$.
- Ⓓ Both Marcus and Shelly are correct because $0.\bar{5}$ is equivalent to both $\frac{5}{10}$ and $\frac{5}{9}$.

- 17.** Josiah records the time, in minutes (min), after he begins to fill his fish tank and the amount of water, in gallons (gal.), in the tank as he fills it. He finds the relationship to be proportional.

Which table could represent Josiah's data?

Josiah's Fish Tank

Ⓐ

Water in Tank (gal.)	2	6	12
Time After Start (min)	8	12	16

Josiah's Fish Tank

Ⓑ

Water in Tank (gal.)	3	7	11
Time After Start (min)	5	20	35

Josiah's Fish Tank

Ⓒ

Water in Tank (gal.)	4	12	20
Time After Start (min)	3	9	15

Josiah's Fish Tank

Ⓓ

Water in Tank (gal.)	5	10	20
Time After Start (min)	4	12	36

- 18.** A factory manufactures 12,000 light bulbs per day. In a random sample of 200 light bulbs, 3 are broken.

Based on the sample, how many light bulbs should the factory expect to be broken each day?

- Ⓐ 3
- Ⓑ 60
- Ⓒ 180
- Ⓓ 203

- 19.** Bobby sells 3-pound bags of peaches. He sells each 3-pound bag for \$7.50 and wants to earn \$900.

If he sells 45 pounds of peaches per hour, how many hours will it take Bobby to earn \$900?

←
→
↶
↷
✖
✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Grade 7 FAST Mathematics

20. How many pounds are in 9 kilograms? Round your answer to the nearest tenth.

←

→

↶

↷

✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

- 21.** An elevator is below the first floor of a hotel at an elevation of -10.42 feet. The elevator travels to the top floor, which has an elevation of 156.53 feet.

What is the change in elevation, in feet, of the elevator?

- Ⓐ 15.02
- Ⓑ 146.11
- Ⓒ 166.95
- Ⓓ 1631.04

Grade 7 FAST Mathematics

- 22.** The cost of 1.75 pounds of cheese is \$9.87. The cost of the cheese is proportional to its weight.

What is the cost, in dollars, for each pound of cheese?

\$

					
1	2	3			
4	5	6			
7	8	9			
	0				
.	-	$\frac{\Box}{\Box}$			

23. Six more than n is less than 14.

Which inequality models this situation?

- Ⓐ $6n < 14$
- Ⓑ $6n > 14$
- Ⓒ $n + 6 < 14$
- Ⓓ $n + 6 > 14$

24. A circle has a circumference of 278 inches.

What is the best approximation for the radius, in inches, of the circle?

- Ⓐ 22.1
- Ⓑ 44.2
- Ⓒ 69.5
- Ⓓ 88.5

25. Ester has a bag with different-colored marbles. The quantity of each marble color is shown in the table.

Marbles

Color	Quantity
Orange	10
Purple	8
Green	6

Ester randomly selects a marble from the bag.

What is the probability that Ester will select a green marble?

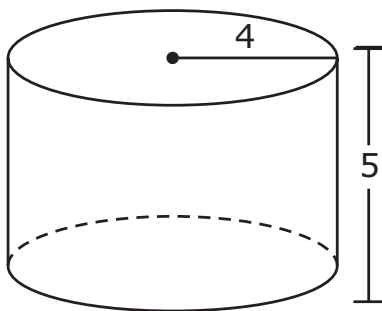
- Ⓐ $\frac{1}{8}$
- Ⓑ $\frac{1}{6}$
- Ⓒ $\frac{1}{4}$
- Ⓓ $\frac{1}{3}$

26. What is the value of $\frac{3^5}{3^4} \cdot 3^3$?

←
→
↶
↷
✖
✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

27. A right circular cylinder, with dimensions in units, is shown.



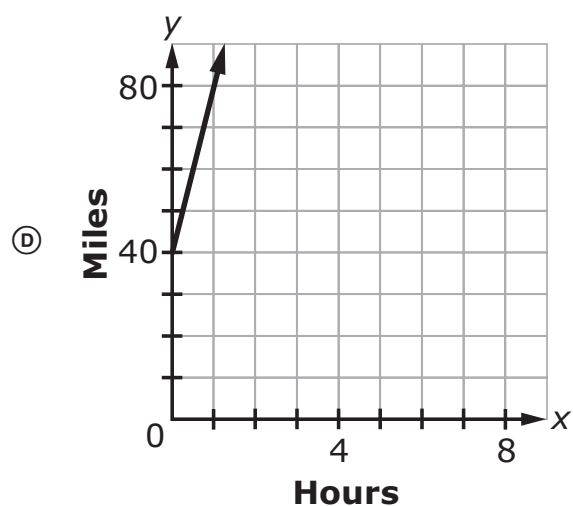
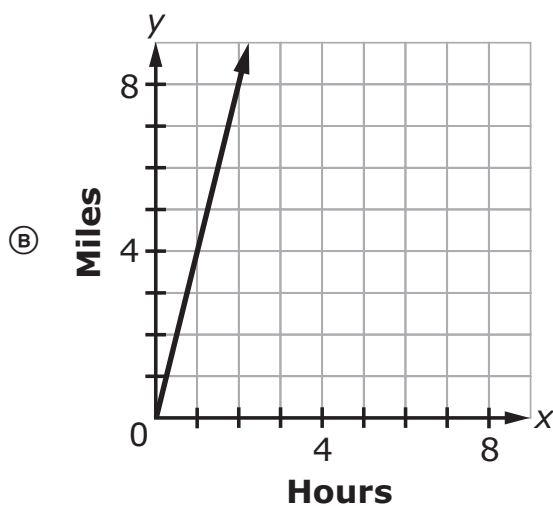
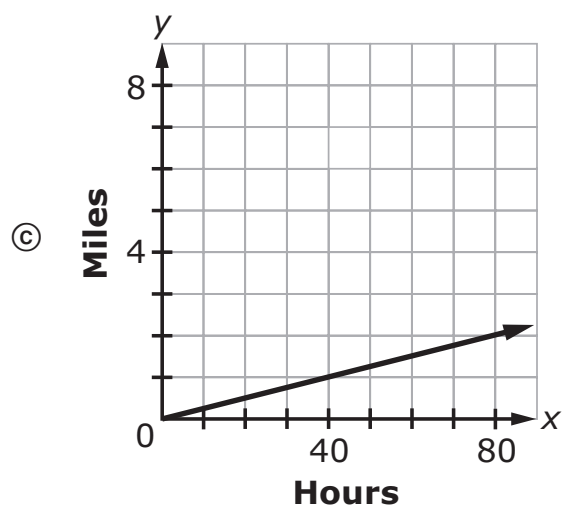
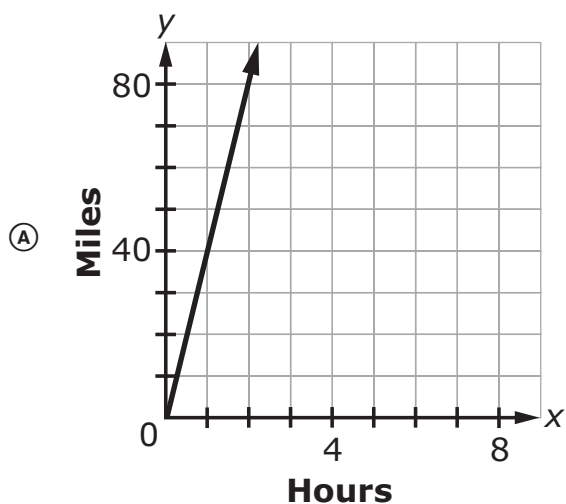
What is the volume, in cubic units, of the cylinder?

- Ⓐ 20π
- Ⓑ 40π
- Ⓒ 80π
- Ⓓ 100π

28. The number of hours driven, x , is proportional to the number of miles traveled, y , as shown in the table.

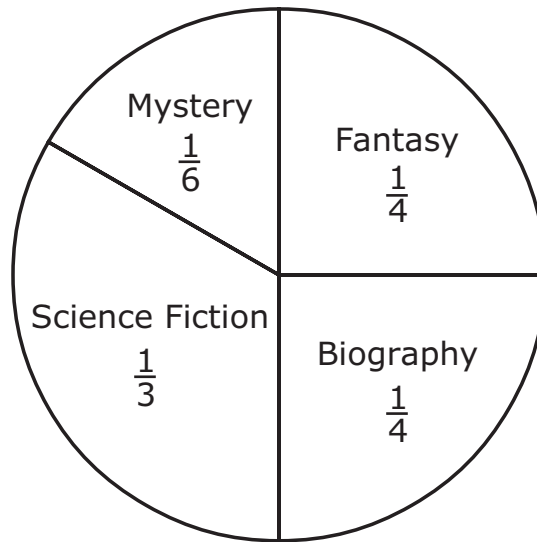
x	0	1	2
y	0	40	80

Which graph represents the proportional relationship?



29. A publishing company surveys 192 students to determine their favorite types of books. The company's data are represented by the graph shown.

Favorite Types of Books



How many students responded that their favorite type of book is mystery?

←

→

↶

↷

✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

- 30.** Jayce and Alberto both play on the basketball team. They record the number of points they each scored in the last 5 games to create the data sets shown.

Jayce: {20, 16, 12, 14, 13}

Alberto: {17, 11, 19, 10, 18}

Complete the statements to describe a comparison of the means of the two data sets. For each blank, fill in the bubble **before** the number or phrase that is correct.

The difference between the means of the data sets is [☐ A 0 ☐ B 1 ☐ C 3 ☐ D 8] .

This indicates that Jayce scored [☐ A fewer points than ☐ B more points than ☐ C the same number of points as] Alberto on average per game.

- 31.** An architect makes a scale drawing of a garage. The garage is 6.2 meters wide. In the scale drawing, the garage is 3.1 centimeters wide.

Enter a value to show the scale the architect uses to make the drawing.

1 centimeter : meters

32. Which table shows a proportional relationship between x and y ?

Ⓐ

x	0	2	4	6
y	0	3	6	9

Ⓑ

x	0	2	4	6
y	5	5	5	5

Ⓒ

x	0	2	4	6
y	1	2	3	4

Ⓓ

x	0	2	4	6
y	0	2	4	8

33. An expression is shown.

$$6 \cdot [(3^3 - 5^2)^2 + 3]$$

What is the value of the expression?

←

→

↶

↷

✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

- 34.** Tory has a deck of cards with different letters on them. The letters are A, E, I, O, U, T, R, and Y. She draws a card at random one time.

Which event can be represented by $P(\text{event}) = 0$?

- Ⓐ drawing a Y
- Ⓑ drawing a vowel
- Ⓒ drawing a number
- Ⓓ drawing a card with a letter from her name

35. What is the product of $\frac{4}{5}$ and -3.1 ?

←

→

↶

↷

✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

36. An expression is shown.

$$3(0.74x - 1.2) + 1.95$$

Which expression is equivalent?

- Ⓐ $2.22x + 0.75$
- Ⓑ $2.22x - 1.65$
- Ⓒ $3.74x - 2.25$
- Ⓓ $3.74x + 3.75$



Grade 7 FAST Mathematics

Test Release: 2026

Grade 7 FAST Mathematics Test Release Support Document

1. An equation of a proportional relationship is shown.

$$y = 2x$$

Which table represents the relationship?

Ⓐ

x	0	2	4
y	0	1	2

Ⓒ

x	0	1	2
y	2	4	6

Ⓑ

x	0	1	2
y	0	2	4

Ⓓ

x	0	1	2
y	2	4	8

Answer Key: B

Percentage of Students Answering Correctly: 34%

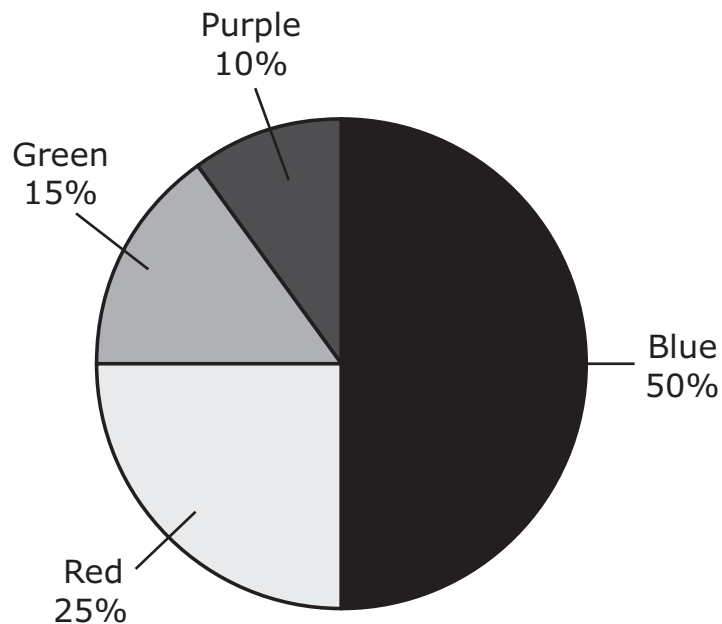
Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.4.4

Benchmark Description: Given any representation of a proportional relationship, translate the representation to a written description, table or equation.

2. In Xander's school club, students vote for their favorite colors. The students' data are shown in the circle graph.

Students' Favorite Colors



Xander knows that 6 students voted for green.

What is the total number of students who voted?

- Ⓐ 15
- Ⓑ 20
- Ⓒ 34
- Ⓓ 40

Answer Key: D

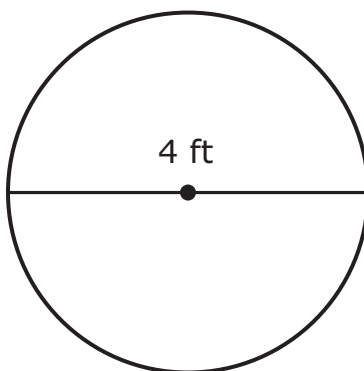
Percentage of Students Answering Correctly: 47%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.1.4

Benchmark Description: Use proportional reasoning to construct, display and interpret data in circle graphs.

3. A circle has a diameter of 4 feet (ft), as shown.



What is the circumference, in feet, of the circle? Round your answer to the nearest tenth.

12.6

←
→
↶
↷
✖
✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 12.6, or any value between 12.56 and 12.6, inclusive

Percentage of Students Answering Correctly: 31%

Reporting Category: Geometric Reasoning

Benchmark: MA.7.GR.1.3

Benchmark Description: Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

4. An expression is shown.

$$\frac{(7^3)^2 \cdot (3^2)^3}{(3 \cdot 7)^4 \cdot 7^2}$$

What is the value of the expression?

- Ⓐ $\frac{3}{7}$
- Ⓑ 9
- Ⓒ $\frac{81}{7}$
- Ⓓ 63

Answer Key: B

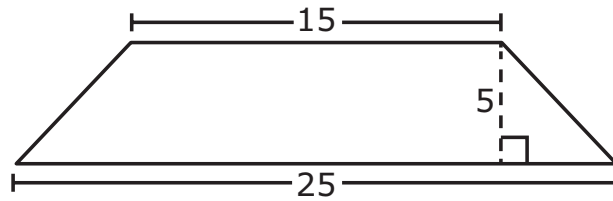
Percentage of Students Answering Correctly: 32%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.NSO.1.1

Benchmark Description: Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.

5. A trapezoid, with dimensions in units, is shown.



What is the area, in square units, of the trapezoid?

- Ⓐ 50
- Ⓑ 100
- Ⓒ 125
- Ⓓ 200

Answer Key: B

Percentage of Students Answering Correctly: 32%

Reporting Category: Geometric Reasoning

Benchmark: MA.7.GR.1.1

Benchmark Description: Apply formulas to find the areas of trapezoids, parallelograms and rhombi.

Grade 7 FAST Mathematics Test Release Support Document

6. Liam rolls a fair 10-sided die. The faces are numbered 1 through 10.

What is the theoretical probability that Liam rolls an 8 or higher? Express your answer as a fraction or a decimal.

The image shows a digital math input interface. At the top, a large rectangular display area shows the fraction $\frac{3}{10}$ in red text. Below this is a toolbar with six circular icons: a left arrow, a right arrow, a circular arrow (undo), a square arrow (redo), a square with an 'X' (clear), and a clipboard with a checkmark. Below the toolbar is a keypad with a grid of buttons. The first four rows of the keypad contain digits 1 through 9, and 0. The fifth row contains a decimal point (.), a negative sign (-), and a fraction template button (represented by a small fraction symbol). The keypad is positioned to the left of a large, empty rectangular area for additional input or calculation.

Answer Key: $\frac{3}{10}$, or any equivalent value

Percentage of Students Answering Correctly: 27%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.2.3

Benchmark Description: Find the theoretical probability of an event related to a simple experiment.

7. An inequality is shown.

$$6 < x + 4$$

What is the solution to the inequality?

- ☒ (A) $x > 2$
- ☐ (B) $x < 2$
- ☐ (C) $x > 10$
- ☐ (D) $x < 10$

Answer Key: A

Percentage of Students Answering Correctly: 34%

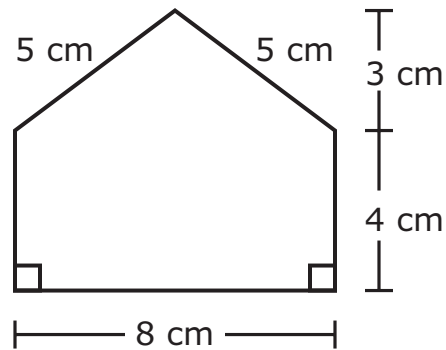
Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.AR.2.1

Benchmark Description: Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

Grade 7 FAST Mathematics Test Release Support Document

8. Jada makes gingerbread cookies in the shape of a house. The dimensions, in centimeters (cm), of one cookie are shown.



What is the area, in square centimeters, of one cookie?

- (A) 39.5
- (B) 44
- (C) 52
- (D) 56

Answer Key: B

Percentage of Students Answering Correctly: 35%

Reporting Category: Geometric Reasoning

Benchmark: MA.7.GR.1.2

Benchmark Description: Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.

Grade 7 FAST Mathematics Test Release Support Document

9. A spinner has five sections that are each a different color. The probability of the spinner arrow stopping on each of the colors is shown.

$$\begin{aligned}P(\text{blue}) &= 0.1 \\P(\text{green}) &= 0.19 \\P(\text{purple}) &= 0.03 \\P(\text{red}) &= 0.18 \\P(\text{yellow}) &= 0.5\end{aligned}$$

Which statement is true?

- Ⓐ It is least likely for the arrow to land on blue.
- Ⓑ It is less likely for the arrow to land on red than on blue.
- Ⓒ It is more likely for the arrow to land on blue than on green.
- Ⓓ It is equally likely and unlikely for the arrow to land on yellow.

Answer Key: D

Percentage of Students Answering Correctly: 33%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.2.2

Benchmark Description: Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.

Grade 7 FAST Mathematics Test Release Support Document

- 10.** A hotel owner increases the price of a room from \$139 per night to \$174 per night.

By about what percentage did the hotel owner increase the price of the room?

- Ⓐ 20%
- Ⓑ 25%
- Ⓒ 35%
- Ⓓ 50%

Answer Key: B

Percentage of Students Answering Correctly: 39%

Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.3.1

Benchmark Description: Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.

Grade 7 FAST Mathematics Test Release Support Document

- 11.** An architect creates a drawing in which a rectangular deck is 11 inches long and 6 inches wide. The drawing is at a scale of 1 inch = 2 feet.

What is the actual area, in square feet, of the deck?

264

←

→

↶

↷

✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Answer Key: 264, or any equivalent value

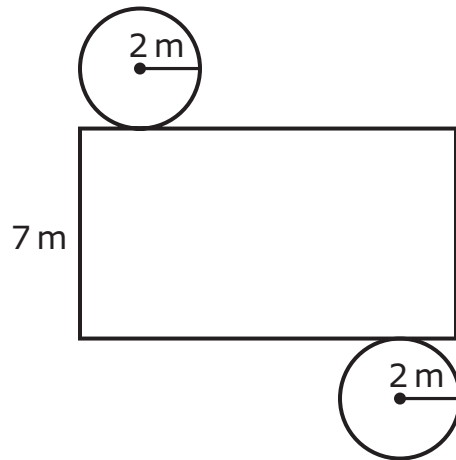
Percentage of Students Answering Correctly: 29%

Reporting Category: Geometric Reasoning

Benchmark: MA.7.GR.1.5

Benchmark Description: Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

- 12.** The net of a right circular cylinder is shown, with a radius of 2 meters (m) and a height of 7 meters.



Which expression can be used to find the surface area, in square meters, of the cylinder?

- Ⓐ $\pi(2)^2 + 2\pi(2)(7)$
- Ⓑ $\pi(4)^2 + 2\pi(4)(7)$
- Ⓒ $2\pi(2)^2 + 2\pi(2)(7)$
- Ⓓ $2\pi(4)^2 + 2\pi(4)(7)$

Answer Key: C

Percentage of Students Answering Correctly: 43%

Reporting Category: Geometric Reasoning

Benchmark: MA.7.GR.2.1

Benchmark Description: Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.

13. Which expression is equivalent to $2.4(7.5 + 10p) - 15p$?

- Ⓐ $27p$
- Ⓑ $18 - 5p$
- Ⓒ $18 + 9p$
- Ⓓ $18 - 12p$

Answer Key: C

Percentage of Students Answering Correctly: 38%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.AR.1.2

Benchmark Description: Determine whether two linear expressions are equivalent.

Grade 7 FAST Mathematics Test Release Support Document

14. Jon's music player picks 100 songs at random. He records the genres of the songs in the table shown.

Genre	Number of Songs
Country	23
Hip-Hop	13
Pop	27
Rock	37

Based on the table, what is the experimental probability that the music player will pick a rock song? Express your answer as a fraction or a decimal.

0.37

←

→

↶

↷

✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 0.37, or any equivalent value

Percentage of Students Answering Correctly: 43%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.2.4

Benchmark Description: Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

15. Six more than one-half of x is equal to 16.

Which equation models this description?

Ⓐ $\frac{1}{2}x + 6 = 16$

Ⓑ $6x + \frac{1}{2} = 16$

Ⓒ $\left(\frac{1}{2} + 6\right)x = 16$

Ⓓ $6\left(x + \frac{1}{2}\right) = 16$

Answer Key: A

Percentage of Students Answering Correctly: 36%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.AR.2.2

Benchmark Description: Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

16. Marcus and Shelly both rewrite $1.\bar{5}$ as a fraction.

- Marcus claims $1.\bar{5} = \frac{15}{10}$ because $1.\bar{5} = 1 + \frac{5}{10} = \frac{15}{10}$.
- Shelly claims $1.\bar{5} = \frac{14}{9}$ because $1.\bar{5} = 1 + \frac{5}{9} = \frac{14}{9}$.

Which statement describes who is correct?

- Ⓐ Shelly is correct because $0.\bar{5}$ is equivalent to $\frac{5}{9}$.
- Ⓑ Marcus is correct because $0.\bar{5}$ is equivalent to $\frac{5}{10}$.
- Ⓒ Neither Marcus nor Shelly is correct because $0.\bar{5}$ is not equivalent to $\frac{5}{10}$ or $\frac{5}{9}$.
- Ⓓ Both Marcus and Shelly are correct because $0.\bar{5}$ is equivalent to both $\frac{5}{10}$ and $\frac{5}{9}$.

Answer Key: A

Percentage of Students Answering Correctly: 30%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.NSO.1.2

Benchmark Description: Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals and percentages to solve mathematical and real-world problems.

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- 17.** Josiah records the time, in minutes (min), after he begins to fill his fish tank and the amount of water, in gallons (gal.), in the tank as he fills it. He finds the relationship to be proportional.

Which table could represent Josiah's data?

Josiah's Fish Tank

Ⓐ

Water in Tank (gal.)	2	6	12
Time After Start (min)	8	12	16

Josiah's Fish Tank

Ⓑ

Water in Tank (gal.)	3	7	11
Time After Start (min)	5	20	35

Josiah's Fish Tank

Ⓒ

Water in Tank (gal.)	4	12	20
Time After Start (min)	3	9	15

Josiah's Fish Tank

Ⓓ

Water in Tank (gal.)	5	10	20
Time After Start (min)	4	12	36

Answer Key: C

Percentage of Students Answering Correctly: 45%

Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.4.1

Benchmark Description: Determine whether two quantities have a proportional relationship by examining a table, graph or written description.

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- 18.** A factory manufactures 12,000 light bulbs per day. In a random sample of 200 light bulbs, 3 are broken.

Based on the sample, how many light bulbs should the factory expect to be broken each day?

- Ⓐ 3
- Ⓑ 60
- Ⓒ 180
- Ⓓ 203

Answer Key: C

Percentage of Students Answering Correctly: 49%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.1.3

Benchmark Description: Given categorical data from a random sample, use proportional relationships to make predictions about a population.

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- 19.** Bobby sells 3-pound bags of peaches. He sells each 3-pound bag for \$7.50 and wants to earn \$900.

If he sells 45 pounds of peaches per hour, how many hours will it take Bobby to earn \$900?

8					
←	→	↶	↷	✖	✓
1	2	3			
4	5	6			
7	8	9			
	0				
.	-	$\frac{\square}{\square}$			

Answer Key: 8, or any equivalent value

Percentage of Students Answering Correctly: 21%

Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.4.5

Benchmark Description: Solve real-world problems involving proportional relationships.

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20. How many pounds are in 9 kilograms? Round your answer to the nearest tenth.

19.8

←

→

↶

↷

✖

✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Answer Key: 19.8, or any value between 19.8 and 19.836, inclusive

Percentage of Students Answering Correctly: 28%

Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.3.3

Benchmark Description: Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

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- 21.** An elevator is below the first floor of a hotel at an elevation of -10.42 feet. The elevator travels to the top floor, which has an elevation of 156.53 feet.

What is the change in elevation, in feet, of the elevator?

- Ⓐ 15.02
- Ⓑ 146.11
- Ⓒ 166.95
- Ⓓ 1631.04

Answer Key: C

Percentage of Students Answering Correctly: 45%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.NSO.2.3

Benchmark Description: Solve real-world problems involving any of the four operations with rational numbers.

Grade 7 FAST Mathematics Test Release Support Document

- 22.** The cost of 1.75 pounds of cheese is \$9.87. The cost of the cheese is proportional to its weight.

What is the cost, in dollars, for each pound of cheese?

\$

←	→	↶	↷	✖	✓
1	2	3			
4	5	6			
7	8	9			
	0				
.	-	$\frac{\Box}{\Box}$			

Answer Key: \$5.64, or any equivalent non-fractional value

Percentage of Students Answering Correctly: 36%

Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.4.2

Benchmark Description: Determine the constant of proportionality within a mathematical or real-world context given a table, graph or written description of a proportional relationship.

23. Six more than n is less than 14.

Which inequality models this situation?

Ⓐ $6n < 14$

Ⓑ $6n > 14$

Ⓒ $n + 6 < 14$

Ⓓ $n + 6 > 14$

Answer Key: C

Percentage of Students Answering Correctly: 45%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.AR.2.1

Benchmark Description: Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

24. A circle has a circumference of 278 inches.

What is the best approximation for the radius, in inches, of the circle?

- Ⓐ 22.1
- Ⓑ 44.2
- Ⓒ 69.5
- Ⓓ 88.5

Answer Key: B

Percentage of Students Answering Correctly: 42%

Reporting Category: Geometric Reasoning

Benchmark: MA.7.GR.1.3

Benchmark Description: Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

25. Ester has a bag with different-colored marbles. The quantity of each marble color is shown in the table.

Marbles

Color	Quantity
Orange	10
Purple	8
Green	6

Ester randomly selects a marble from the bag.

What is the probability that Ester will select a green marble?

- (A) $\frac{1}{8}$
- (B) $\frac{1}{6}$
- (C) $\frac{1}{4}$
- (D) $\frac{1}{3}$

Answer Key: C

Percentage of Students Answering Correctly: 38%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.2.3

Benchmark Description: Find the theoretical probability of an event related to a simple experiment.

26. What is the value of $\frac{3^5}{3^4} \cdot 3^3$?

81

←
→
↶
↷
✖
✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Answer Key: 81, or any equivalent value

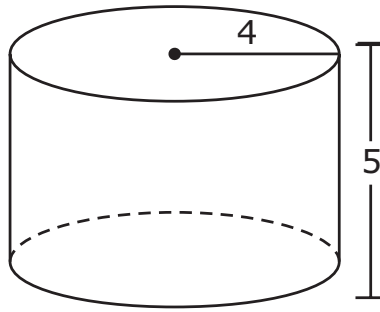
Percentage of Students Answering Correctly: 20%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.NSO.1.1

Benchmark Description: Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.

27. A right circular cylinder, with dimensions in units, is shown.



What is the volume, in cubic units, of the cylinder?

- Ⓐ 20π
- Ⓑ 40π
- Ⓒ 80π
- Ⓓ 100π

Answer Key: C

Percentage of Students Answering Correctly: 24%

Reporting Category: Geometric Reasoning

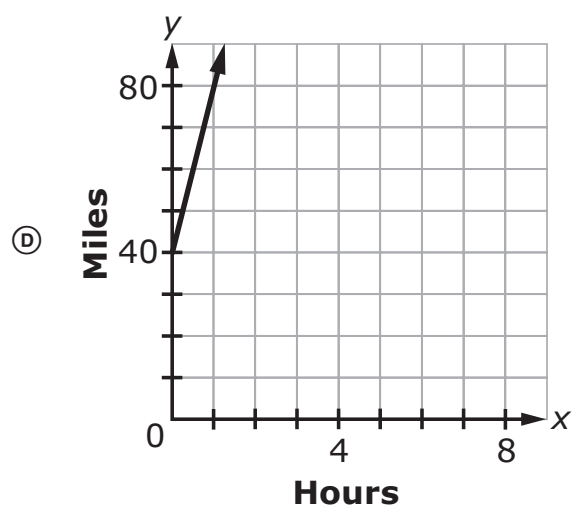
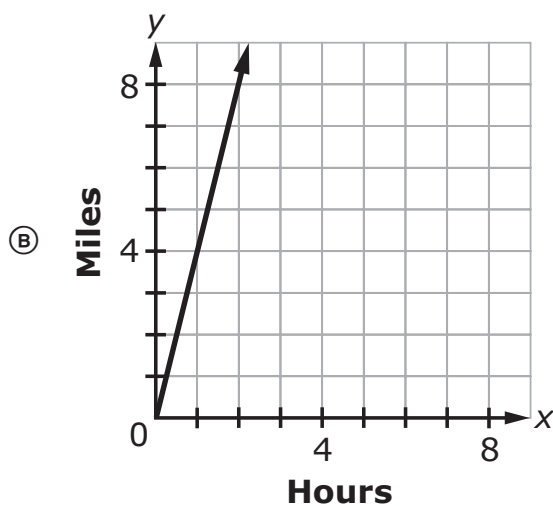
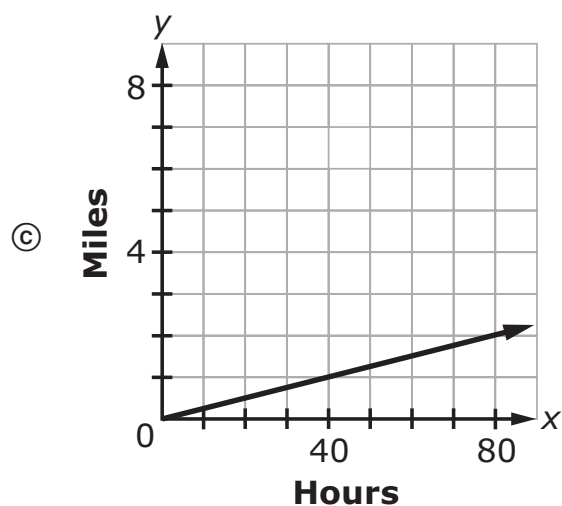
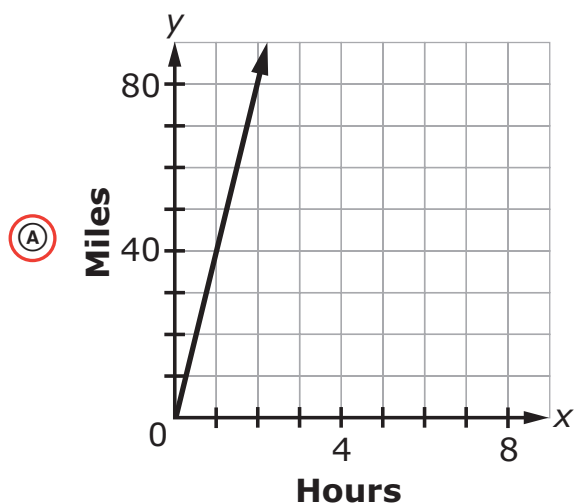
Benchmark: MA.7.GR.2.3

Benchmark Description: Solve mathematical and real-world problems involving volume of right circular cylinders.

28. The number of hours driven, x , is proportional to the number of miles traveled, y , as shown in the table.

x	0	1	2
y	0	40	80

Which graph represents the proportional relationship?



Grade 7 FAST Mathematics Test Release Support Document

Answer Key: A

Percentage of Students Answering Correctly: 63%

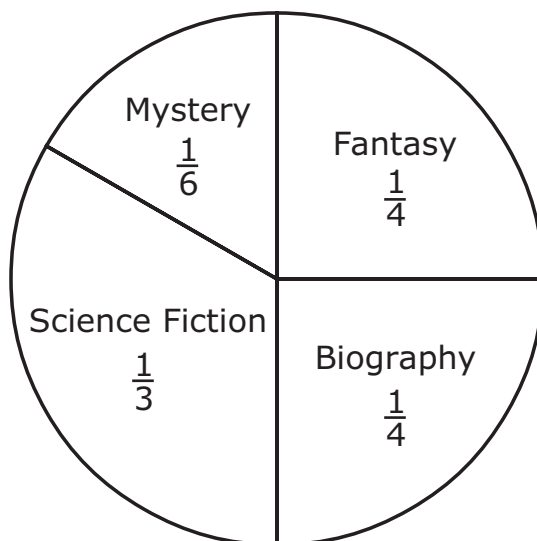
Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.4.3

Benchmark Description: Given a mathematical or real-world context, graph proportional relationships from a table, equation or a written description.

29. A publishing company surveys 192 students to determine their favorite types of books. The company's data are represented by the graph shown.

Favorite Types of Books



How many students responded that their favorite type of book is mystery?

32

←
→
↶
↷
✖
✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Grade 7 FAST Mathematics Test Release Support Document

Answer Key: 32, or any equivalent value

Percentage of Students Answering Correctly: 32%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.1.4

Benchmark Description: Use proportional reasoning to construct, display and interpret data in circle graphs.

Grade 7 FAST Mathematics Test Release Support Document

- 30.** Jayce and Alberto both play on the basketball team. They record the number of points they each scored in the last 5 games to create the data sets shown.

Jayce: {20, 16, 12, 14, 13}

Alberto: {17, 11, 19, 10, 18}

Complete the statements to describe a comparison of the means of the two data sets. For each blank, fill in the bubble **before** the number or phrase that is correct.

The difference between the means of the data sets is [☒ A 0 ☐ B 1 ☐ C 3 ☐ D 8] .

This indicates that Jayce scored [☒ A fewer points than ☐ B more points than ☐ C the same number of points as] Alberto on average per game.

Answer Key: A, C

Percentage of Students Answering Correctly: 36%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.1.2

Benchmark Description: Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results and draw conclusions about the two populations.

Grade 7 FAST Mathematics Test Release Support Document

- 31.** An architect makes a scale drawing of a garage. The garage is 6.2 meters wide. In the scale drawing, the garage is 3.1 centimeters wide.

Enter a value to show the scale the architect uses to make the drawing.

1 centimeter :

2

meters

Answer Key: 2, or any equivalent value

Percentage of Students Answering Correctly: 39%

Reporting Category: Geometric Reasoning

Benchmark: MA.7.GR.1.5

Benchmark Description: Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

32. Which table shows a proportional relationship between x and y ?

Ⓐ

x	0	2	4	6
y	0	3	6	9

Ⓑ

x	0	2	4	6
y	5	5	5	5

Ⓒ

x	0	2	4	6
y	1	2	3	4

Ⓓ

x	0	2	4	6
y	0	2	4	8

Answer Key: A

Percentage of Students Answering Correctly: 53%

Reporting Category: Proportional Reasoning and Relationships

Benchmark: MA.7.AR.4.1

Benchmark Description: Determine whether two quantities have a proportional relationship by examining a table, graph or written description.

33. An expression is shown.

$$6 \cdot [(3^3 - 5^2)^2 + 3]$$

What is the value of the expression?

42

← → ↶ ↷ ✖ ✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Answer Key: 42, or any equivalent value

Percentage of Students Answering Correctly: 27%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.NSO.2.1

Benchmark Description: Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value.

Grade 7 FAST Mathematics Test Release Support Document

- 34.** Tory has a deck of cards with different letters on them. The letters are A, E, I, O, U, T, R, and Y. She draws a card at random one time.

Which event can be represented by $P(\text{event}) = 0$?

- Ⓐ drawing a Y
- Ⓑ drawing a vowel
- Ⓒ drawing a number
- Ⓓ drawing a card with a letter from her name

Answer Key: C

Percentage of Students Answering Correctly: 51%

Reporting Category: Data Analysis and Probability

Benchmark: MA.7.DP.2.2

Benchmark Description: Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.

35. What is the product of $\frac{4}{5}$ and -3.1 ?

-2.48

←
→
↶
↷
✖
✓

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Answer Key: -2.48, or any equivalent value

Percentage of Students Answering Correctly: 37%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.NSO.2.2

Benchmark Description: Add, subtract, multiply and divide rational numbers with procedural fluency.

36. An expression is shown.

$$3(0.74x - 1.2) + 1.95$$

Which expression is equivalent?

- Ⓐ $2.22x + 0.75$
- Ⓑ $2.22x - 1.65$
- Ⓒ $3.74x - 2.25$
- Ⓓ $3.74x + 3.75$

Answer Key: B

Percentage of Students Answering Correctly: 47%

Reporting Category: Number Sense and Operations and Algebraic Reasoning

Benchmark: MA.7.AR.1.2

Benchmark Description: Determine whether two linear expressions are equivalent.